Log Book Entry 4.1

NB2420 Electronic Instrumentation · Manual 4.1 · Opamp basics

Hand in via BrightSpace

Name(s): Partner:

Date: Group:

PREPARATION

complete before the practicum

1. *What theories, laws, and/or formulas are most important for this experiment?*
2. *What do you expect to learn?*
3. *What do you expect to observe, and why?*

THREE CONFIGURATIONS

Tasks A1, A3, A5, I1, I4, I5, C1

The side-by-side view is the point. Each column tells you when to reach for that topology.

Property	Non-inverting	Inverting	Transimpedance
Gain formula			
Input variable (V or I)			
Input impedance			
Output impedance			
Phase U_{out} vs U_{in}			
Typical use case			

Predicted, simulated, measured gain Task C2

Circuit	Predicted G	Simulated G	Measured G	Discrepancy
Non-inverting, $R_a/R_b =$ 3.3k /1k				
Non-inverting, $R_a/R_b =$ 1k /3.3k				
Inverting, $R_f/R_i =$ 1k /220				

Buffer and transimpedance Tasks I3 and I5

Source voltage without the buffer:

Source voltage with the buffer:

Photodiode U_{out}, LED off and on:

R_f used, and the current it implies:

Saturation Tasks S1 and S3

Input amplitude where the output first clipped:

Clipping levels seen, positive and negative:

I_{in} where the transimpedance stage saturated:

CONCLUSIONS

Tasks C3 and C4

C3 When do you reach for a buffer? *Use your Task I3 numbers. A gain of 1 sounds useless, so say what it actually buys you.*

C4 A one-line design rule for saturation *Given a supply and a desired output swing, what is the largest input you may apply?*

Graph *Plot U_{out} against U_{in} for one amplifier, or U_{out} against LED drive for the photodiode, and mark where it stops being a straight line.*

REFLECTION

complete before submission

1. *Which results matched or differed from your expectations, and why?*
2. *What is the most important thing you learned?*
3. *What would you do differently next time?*

TA check-off: Date: